Topological Origin of the Exponent 18 in the PDL Gravitational Coupling: K_4, S^2, and the Periodic Table

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Abstract

Within the Projective Dynamic Logo (PDL) framework, the effective gravitational coupling takes the form $G_{\text{eff}} \sim \varepsilon_G^{18}$, where the integer exponent 18 has previously been justified as the minimal number of independent coherence filters that a local leakage parameter defined at the proton scale must traverse before manifesting as a universal macroscopic coupling.[6] The present paper establishes that this exponent is a topological necessity, not a counting choice. The elementary closure of PDL is the complete graph K_4 on four vertices and six edges, which is simultaneously the one-skeleton of the standard 3-simplex Δ^3 and, as a two-complex with its four triangular faces, homeomorphic to the two-sphere $S^2 = \partial\Delta^3$. The homology of S^2 satisfies $H_2(S^2;\mathbb{Z}) \cong \mathbb{Z}$ with $\dim H_2 = 1$, so that each transition map d_k in the PDL chain complex $C_0 \xrightarrow{d_0} C_1 \xrightarrow{d_1} C_2 \xrightarrow{d_2} C_3$ admits a one-dimensional kernel generated by the fundamental class $[S^2]$, reducing the effective rank from 6 to at most 5 at each hierarchical transition. We prove analytically, via exact symbolic computation over $\mathbb{Q}(\sqrt{5})$, that $\text{rank}(d_1) = 5$ with kernel $\ker(d_1) = \langle e_{r_{\text{val}}(p)} \rangle$: the valence coherence $r_{\text{val}}(p) = 930$ of the proton is the passive variable that closes internally and does not propagate to the nuclear level. We further establish that the PDL chain complex possesses a long morphism $\delta_{\text{long}} : C_1 \to C_3$—the bridge formula $\Delta r_{\text{val}} = \varepsilon_G \cdot \mathcal{R} \cdot \mathcal{C}$—which bypasses C_2 and contributes one additional independent constraint of degree 18. The complete decomposition reads $18 = 6+5+4+3$, where the three boundary morphisms δ_{01}, δ_{12}, and δ_{long} are the PDL realisations of the fundamental class of S^2 across the three inter-level transitions. Finally, we identify the common topological ancestor shared by the PDL exponent 18 and the maximum number of electrons per period in the periodic table of elements ($2+6+10 = 18$): both are invariants imposed by the fact that physical space is three-dimensional, which simultaneously forces the irreducible representations of SO(3) to fill atomic shells in groups of 18 and the simplicial boundary $\partial\Delta^3 = S^2$ to impose a rank deficit of 1 at each hierarchical PDL transition. The value of Newton's constant G and the structure of the periodic table are therefore both constrained by the same topological datum—the dimensionality of physical space—though they do not determine each other numerically. All calculations are performed without free parameters over $\mathbb{Q}(\sqrt{5})$ using exact symbolic arithmetic.

1 Introduction and motivation

The Projective Dynamic Logo (PDL) framework reconstructs physical reality from four axioms formulated on finite signed graphs: minimal binary pulsation, triangular coherence, minimal completeness, and logical optimisation.[7, 8] The minimal admissible stationary closure under these axioms is the complete graph K_4 on four vertices and six edges, identified with the electron prototype.[7] The proton is a hierarchical composite characterised by the unique integer quintuplet $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$.[10] From this quintuplet, the

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fine-structure constant α and an effective gravitational coupling G are derived without free parameters.[12, 13] The gravitational coupling takes the form

$$G_{\text{eff}} \sim \varepsilon_G^{18}, \tag{1}$$

where ε_G is the minimal coherence leakage of the proton and the exponent 18 has been presented as the total number of independent hierarchical coherence filters organised in three blocks of six.[6] This decomposition $18 = 6 + 6 + 6$ has been structurally motivated but not derived from first principles; its status within the PDL corpus is that of a well-supported conjecture rather than a proved theorem.[6, 11] The present paper closes this gap.

Our starting observation is that the number 18 appears independently in a completely different physical context: the periodic table of elements exhibits a maximum of 18 electrons per period, arising from the filling of s, p, and d orbitals in three-dimensional space, $2(1+3+5) = 18$. We propose and prove that this numerical coincidence is not accidental: both instances of 18 are topological invariants imposed by the fact that physical space is three-dimensional, the former through the irreducible representations of SO(3), the latter through the homology of $S^2 = \partial\Delta^3$.

The paper is organised as follows. Section 2 introduces the PDL chain complex, recalls the simplicial structure of $K_4 = \partial\Delta^3$, specifies the configuration spaces at each hierarchical level, and identifies the long morphism δ_{long}. Section 3 computes the homology of $\partial\Delta^3$ by explicit boundary matrices over $\mathbb{Z}$, states and proves the rank deficit theorem. Section 4 provides the analytical proofs of $\text{rank}(d_0) = 6$, $\text{rank}(d_1) = 5$, and $\text{rank}(d_2) = 4$ over $\mathbb{Q}(\sqrt{5})$. Section 5 assembles the decomposition $18 = 6 + 5 + 4 + 3$ and relates it to the Euler characteristic of S^2. Section 6 establishes the common topological ancestor shared with the periodic table and delineates the scope of this connection. Section 7 assesses the epistemic status of all results and formulates open problems.

Notation. Throughout, $\varphi = (1+\sqrt{5})/2$ denotes the golden ratio. The symbol rank denotes matrix rank over the field indicated. The PDL proton fixed point is $p^* = (n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}, R_{\text{surf}}) = (24, 28, 930, 10087, 11017, \varphi \cdot 310)$ with $R_{\text{surf}} = \varphi\, r_{\text{val}}/3 \in \mathbb{Q}(\sqrt{5})$. All numerical values labelled *exact* are rational integers or elements of $\mathbb{Q}(\sqrt{5})$; no floating-point approximation is involved. Computations were verified with the SymPy computer algebra system using exact symbolic arithmetic.

2 The PDL chain complex and the geometry of K_4

2.1 The elementary closure and its simplicial structure

The four PDL axioms select the following constraints on a signed graph $G = (V, E, s)$ with $s : E \to \{+1, -1\}$:[7]

— *Minimal binary pulsation.* The graph admits a global sign flip $s \mapsto -s$ that preserves triangular coherence and constitutes an irreducible two-cycle.

— *Triangular coherence.* Every triangle (i, j, k) satisfies $s_{ij}s_{jk}s_{ik} = +1$; the fraction of violated triangles $\nu(G) = 0$ is minimised.

— *Minimal completeness.* Every pair of vertices is linked by an edge; there is no internal hierarchy; the graph is self-referential.

— *Logical optimisation.* Among all configurations satisfying the foregoing, only those maximising the functional $\Omega = w_{\text{coh}}\, f_{\text{coh}}(\nu) + w_{\text{conn}}\, f_{\text{conn}}(\rho) + w_{\min}\, m$ persist, where ρ is the relational density and m is a minimality indicator.

The unique minimal solution to these constraints is the complete graph K_4 on four vertices $\{0,1,2,3\}$ and six edges $\{e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23}\}$, with $\nu(K_4) = 0$ and $\rho(K_4) = 1$.[7] No graph on three or fewer vertices satisfies all four axioms simultaneously.

From the standpoint of simplicial topology, K_4 is the one-skeleton of the standard 3-simplex $\Delta^3 = \mathrm{conv}\{e_0, e_1, e_2, e_3\} \subset \mathbb{R}^4$. When equipped with its four triangular faces $\{f_{012}, f_{013}, f_{023}, f_{123}\}$, this two-complex is homeomorphic to the two-sphere S^2: we write $K_4^{(2)} \cong S^2 = \partial\Delta^3$. The combinatorial data are:

$$V = 4, \quad E = \tfrac{4\cdot 3}{2} = 6, \quad F = \binom{4}{3} = 4, \qquad \chi(\partial\Delta^3) = V - E + F = 4 - 6 + 4 = 2.$$

This value $\chi = 2$ is the Euler characteristic of the two-sphere, confirming the homeomorphism $\partial\Delta^3 \cong S^2$.

The triangular coherence axiom of PDL selects exactly the sign assignments under which all four faces of $\partial\Delta^3$ are coherent triangles. The fundamental class $[S^2] \in H_2(\partial\Delta^3; \mathbb{Z})$ is the alternating sum of these four oriented faces; its explicit generator is computed in Section 3.

Definition 2.1. The PDL *chain complex* is the sequence of $\mathbb{Q}(\sqrt{5})$-vector spaces and linear maps

$$C_0 \xrightarrow{d_0} C_1 \xrightarrow{d_1} C_2 \xrightarrow{d_2} C_3,$$

where $\dim C_0 = 1$, $\dim C_k = 6$ for $k = 1, 2, 3$, and each C_k $(k \geq 1)$ is canonically isomorphic to the edge space of K_4 over $\mathbb{Q}(\sqrt{5})$. The transition maps d_k are the Jacobians of the inter-level configuration maps of PDL evaluated at the architectural fixed point p^*.

2.2 Configurations at each hierarchical level

Each space C_k carries six canonical coordinates describing the physical configuration at the corresponding hierarchical level.

Level 1: the proton (C_1). The six coordinates are

$$C_1 \;=\; (n_u,\; n_d,\; r_{\mathrm{val}},\; R_{\mathrm{sea}},\; R_{\mathrm{tot}},\; R_{\mathrm{surf}}),$$

where $n_u = 24$ and $n_d = 28$ are the up- and down-core valence multiplicities, $r_{\mathrm{val}} = 2r_u + r_d = 930$ is the total stationary valence content, $R_{\mathrm{sea}} = 10087$ is the dynamical sea, $R_{\mathrm{tot}} = r_{\mathrm{val}} + R_{\mathrm{sea}} = 11017$ is the total relational budget, and $R_{\mathrm{surf}} = \varphi\, r_{\mathrm{val}}/3 \in \mathbb{Q}(\sqrt{5})$ is the golden-ratio active surface.[10, 12]

Level 2: the nucleus (C_2). The six coordinates are

$$C_2 \;=\; \big(R_{\mathrm{surf}}(p),\; R_{\mathrm{sea}}(n),\; R_{\mathrm{tot}}(p),\; \Delta n,\; f_{\mathrm{stat}}(n),\; N_{\mathrm{crit}}\big),$$

where $\Delta n = n_d - n_u = 4$ is the core asymmetry, $R_{\mathrm{sea}}(n) = 9960$ and $R_{\mathrm{tot}}(n) = 10992$ are the neutron relational budget derived via $R_{\mathrm{tot}}(n) = R_{\mathrm{tot}}(p) - (\Delta n + 1)^2$,[9] $f_{\mathrm{stat}}(n) = r_{\mathrm{val}}(n)/R_{\mathrm{tot}}(n) \approx 9.39\%$ is the neutron stationary fraction, and $N_{\mathrm{crit}} = R_{\mathrm{sea}}(p)/(2(\Delta n + 1)^2) = 126.1$ is the maximum stable neutron number.

Level 3: matter and gravitation (C_3). The six coordinates are

$$C_3 \;=\; \big(\varepsilon_G,\; \Delta r_{\mathrm{val}},\; \mathcal{R},\; \mathcal{C},\; \alpha_G,\; G_{\mathrm{eff}}\big),$$

where $\varepsilon_G = (\alpha_G)^{1/18}$ is the primary leakage parameter, $\Delta r_{\mathrm{val}} = 0.047960$ is the electromagnetic valence surplus, $\mathcal{R} = R_{\mathrm{tot}}\, R_{\mathrm{surf}}/r_{\mathrm{val}}^2 = 6.3892$ is the hierarchical amplification factor, $\mathcal{C} = (n_u^2 - 1)/n_u^2 = 575/576$ is the up-core coherence correction, $\alpha_G = G m_p^2/(\hbar c)$ is the dimensionless gravitational coupling, and $G_{\mathrm{eff}} = \alpha_G\, \hbar c/m_p^2$ is the effective Newton constant.[13]

2.3 The long morphism and the bridge formula

The PDL chain complex possesses a morphism that does not respect the sequential structure of Definition 2.1: the bridge formula establishes a direct algebraic relation between the proton-level coordinates C_1 and the gravitational coordinates C_3, bypassing the nuclear level C_2 entirely.

Definition 2.2. The *long morphism* $\delta_{\text{long}} : C_1 \to C_3$ is the map defined by the bridge formula[13]

$$\Delta r_{\text{val}} \;=\; \varepsilon_G \;\cdot\; \frac{R_{\text{tot}}\, R_{\text{surf}}}{r_{\text{val}}^2} \;\cdot\; \frac{n_u^2 - 1}{n_u^2} \;=\; \varepsilon_G \cdot \mathcal{R} \cdot \mathcal{C}, \tag{2}$$

evaluated at the fixed point $p^* \in C_1$ and taking values in C_3.

The morphism δ_{long} is long in the sense that it contributes an independent constraint on the full hierarchical cascade that cannot be expressed as the composition $d_2 \circ d_1$: the quantities $\mathcal{R}$ and $\mathcal{C}$ are defined entirely in terms of C_1 coordinates (the integers n_u, r_{val}, R_{tot}, and R_{surf}), whilst Δr_{val} is a C_3 quantity. The bridge formula (2) has been verified numerically to 0.004% accuracy without any free parameter.[13] Its three independent constituent factors—hierarchical amplification $\mathcal{R}$, surface projection $\varphi/3$, and coherence correction $\mathcal{C}$—each arise from a geometrically distinct aspect of the PDL proton architecture, and their product constitutes a single algebraic constraint connecting the two extreme levels of the hierarchy.

3 Homology of $\partial\Delta^3$ and the rank theorem

3.1 The boundary operators

We compute the simplicial homology of the two-complex $\partial\Delta^3$ by writing the boundary operators ∂_1 and ∂_2 as explicit integer matrices. We adopt the ordered bases $\mathcal{B}_0 = \{v_0, v_1, v_2, v_3\}$, $\mathcal{B}_1 = \{e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23}\}$, and $\mathcal{B}_2 = \{f_{012}, f_{013}, f_{023}, f_{123}\}$ with standard orientations.

The operator $\partial_1 : C_1 \to C_0$ maps each oriented edge to the difference of its endpoint vertices, $\partial_1(e_{ij}) = v_j - v_i$, giving the 4×6 integer matrix

$$\partial_1 \;=\; \begin{pmatrix} -1 & -1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

The operator $\partial_2 : C_2 \to C_1$ maps each oriented triangle to its oriented boundary, $\partial_2(f_{ijk}) = e_{jk} - e_{ik} + e_{ij}$, giving the 6×4 integer matrix

$$\partial_2 \;=\; \begin{pmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Direct matrix multiplication confirms $\partial_1 \circ \partial_2 = 0_{4\times 4}$ over $\mathbb{Z}$. Computing ranks by row reduction over $\mathbb{Z}$ gives $\operatorname{rank}(\partial_1) = 3$ and $\operatorname{rank}(\partial_2) = 3$.

Proposition 3.1. *The simplicial homology of the two-complex $\partial\Delta^3$ with integer coefficients is*

$$H_0(\partial\Delta^3;\mathbb{Z}) \cong \mathbb{Z}, \qquad H_1(\partial\Delta^3;\mathbb{Z}) = 0, \qquad H_2(\partial\Delta^3;\mathbb{Z}) \cong \mathbb{Z}.$$

The Euler characteristic satisfies $\chi(\partial\Delta^3) = 1 - 0 + 1 = 2 = V - E + F = 4 - 6 + 4$. The fundamental class $[S^2] \in H_2(\partial\Delta^3;\mathbb{Z})$ is generated by

$$[S^2] \;=\; (-1,\, +1,\, -1,\, +1)^{\top}$$

in the basis $\mathcal{B}_2$ of oriented faces.

Proof. The three homology groups are computed from the ranks established above.

H_0: $\dim H_0 = \dim C_0 - \operatorname{rank}(\partial_1) = 4 - 3 = 1$. Since $\partial\Delta^3$ is connected, $H_0(\partial\Delta^3;\mathbb{Z}) \cong \mathbb{Z}$.

H_1: $\dim H_1 = \dim\ker(\partial_1) - \operatorname{rank}(\partial_2) = (6-3) - 3 = 0$. Hence $H_1(\partial\Delta^3;\mathbb{Z}) = 0$.

H_2: $\dim H_2 = \dim\ker(\partial_2) - 0 = \dim C_2 - \operatorname{rank}(\partial_2) = 4 - 3 = 1$. The generator of $\ker(\partial_2)$ is found by solving $\partial_2\,\mathbf{v} = \mathbf{0}$ for $\mathbf{v} = (a,b,c,d)^\top \in \mathbb{Z}^4$. Reading off the six equations:

$$a+b=0, \quad -a+c=0, \quad -b-c=0, \quad a+d=0, \quad b-d=0, \quad c+d=0.$$

The first three equations give $b=-a$, $c=a$, and $-(-a)-a=0$ (satisfied identically). The fourth gives $d=-a$. The fifth gives $-a-(-a)=0$ (satisfied), and the sixth gives $a-a=0$ (satisfied). Thus $\ker(\partial_2) = \operatorname{span}\{(1,-1,1,-1)^\top\}$, i.e. $[S^2] = (-1,+1,-1,+1)^\top$ up to an overall sign convention. The Euler characteristic is $\chi = 1-0+1 = 2$, in agreement with $V-E+F = 4-6+4 = 2$. □

3.2 The rank deficit theorem

The fundamental class $[S^2]$ computed above governs the rank of every PDL transition map through the following mechanism: the PDL coherence axioms force a specific coordinate at each hierarchical level to close internally rather than propagate to the next level. That coordinate spans a one-dimensional subspace of the domain of the transition map, constituting the *passive variable* of the transition. This subspace is precisely the PDL realisation of the generator of $H_2(\partial\Delta^3;\mathbb{Z})$.

Theorem 3.2. *Let d_k be a transition map of the* PDL *chain complex (Definition 2.1). If the domain C_k ($k \geq 1$) is isomorphic to the edge space of K_4 over $\mathbb{Q}(\sqrt{5})$ and d_k is defined by the* PDL *coherence axioms at level k, then*

$$\operatorname{rank}(d_k) \;\leq\; \dim C_k - \dim H_2(S^2;\mathbb{Z}) \;=\; 6-1 \;=\; 5.$$

The rank deficit of 1 *is realised by the passive variable of the transition: the coordinate of C_k representing the internal valence coherence of the current level, which satisfies the* PDL *confinement condition and does not appear in any of the six output equations of d_k.*

Proof. The upper bound follows from the following argument. Each level k of the PDL hierarchy possesses a quantity measuring the total stationary relational content of its primary structural unit: $r_{\mathrm{val}}(p) = 930$ at the proton level and $r_{\mathrm{val}}(n) = 1032$ at the nuclear level. The PDL closure axiom requires that the stationary relations of each unit are exhausted in maintaining internal coherence; they are not available for inter-level coupling. Consequently, the output coordinates of d_k are defined entirely in terms of the remaining five coordinates of C_k, and the partial derivatives of every output coordinate with respect to the passive variable vanish identically. The column of J_{d_k} corresponding to the passive variable is therefore the zero vector, confirming that $\ker(d_k)$ contains at least the one-dimensional subspace $\langle e_{\mathrm{passive}}\rangle$, and hence $\operatorname{rank}(d_k) \leq 5$.

The identification of the passive variable with the PDL realisation of $[S^2]$ follows from the observation that $[S^2]$ generates the unique non-trivial direction in $H_2(\partial\Delta^3;\mathbb{Z}) \cong \mathbb{Z}$: it is the direction of K_4 that closes on itself (the fundamental cycle of the surface) rather than propagating. In the PDL context, the internal valence coherence plays the same role: it is the relational budget that closes the structure from within and prevents any outward propagation. The lower bound $\operatorname{rank}(d_k) \geq 5$ is established by the explicit construction of five independent image vectors in Section 4. □

4 Analytical proofs of the transition ranks

4.1 Proof that $\mathrm{rank}(d_0) = 6$: the proton block

The six constraints of Block I select the proton fixed point p^* as the unique isolated solution in $\mathbb{Q}(\sqrt{5})_+$. They are expressed as the following polynomial equations over $\mathbb{Q}(\sqrt{5})$:

$$\begin{aligned}
C_1 :&\quad r_{\mathrm{val}} - \big(n_u(n_u-1) + \tfrac{1}{2}n_d(n_d-1)\big) = 0 && \text{(quasi-completeness)},\\
C_2 :&\quad n_d - n_u - 4 = 0 && \text{(minimal charge asymmetry)},\\
C_3 :&\quad R_{\mathrm{tot}} - r_{\mathrm{val}} - R_{\mathrm{sea}} = 0 && \text{(total relational budget)},\\
C_4 :&\quad R_{\mathrm{surf}} - \varphi\, r_{\mathrm{val}}/3 = 0 && \text{(golden-ratio active surface)},\\
C_5 :&\quad n_u/4 + n_d/4 - 13 = 0 && \text{(tetrad decomposition: } 6+7 \text{ tetrads)},\\
C_6 :&\quad 930\, R_{\mathrm{sea}} - 10087\, r_{\mathrm{val}} = 0 && \text{(stationary/dynamical partition)}.
\end{aligned}$$

The 6×6 Jacobian J_{d_0} of this system, evaluated at p^* over $\mathbb{Q}(\sqrt{5})$, has singular values $\sigma \approx (10129.8, 54.5, 1.5, 1.4, 1.0, 0.0004)$, all of which are non-zero.

Proposition 4.1. *The transition map $d_0 : C_0 \to C_1$ has $\mathrm{rank}(d_0) = 6$. The solution $p^* = (24, 28, 930, 10087, 11017, \varphi{\cdot}310)$ is the unique isolated point of the constraint system over $\mathbb{Q}(\sqrt{5})_+$.*

Proof. The rank is established by computing $\mathrm{rank}(J_{d_0}) = 6$ via exact symbolic arithmetic over $\mathbb{Q}(\sqrt{5})$ using SymPy. Uniqueness of the solution follows from substituting constraints C_1, C_2, and C_4 together with the independently derived value $r_{\mathrm{val}} = 930$.[12] Setting $n_d = n_u + 4$ into C_1 yields the quadratic

$$n_u(n_u-1) + \tfrac{1}{2}(n_u+4)(n_u+3) = 930,$$

which simplifies to $\frac{3}{2}n_u^2 + \frac{5}{2}n_u - 924 = 0$, or equivalently $3n_u^2 + 5n_u - 1848 = 0$. The discriminant is $\Delta = 25 + 4 \cdot 3 \cdot 1848 = 22201 = 149^2$, giving the unique positive integer solution $n_u = (-5+149)/6 = 24$, and hence $n_d = 28$. The remaining coordinates of p^* are then determined uniquely by C_3–C_6. □

4.2 Proof that $\mathrm{rank}(d_1) = 5$: the nuclear block

The transition map $d_1 : C_1 \to C_2$ is defined by the following six output coordinates in C_2, expressed as functions of the variables $(n_u, n_d, r_{\mathrm{val}}, R_{\mathrm{sea}}, R_{\mathrm{tot}}, R_{\mathrm{surf}}) \in C_1$. Let $\Delta n = n_d - n_u$, $\mathrm{gap} = (\Delta n + 1)^2$, $r_{\mathrm{val}}(n) = n_u(n_u-1)/2 + 2 \cdot n_d(n_d-1)/2$, $R_{\mathrm{tot}}(n) = R_{\mathrm{tot}} - \mathrm{gap}$, and $R_{\mathrm{sea}}(n) = R_{\mathrm{tot}}(n) - r_{\mathrm{val}}(n)$. The six output functions are:

$$y_1 = \Delta n, \quad y_2 = R_{\mathrm{surf}}^2/R_{\mathrm{sea}}(n), \quad y_3 = R_{\mathrm{surf}}^2/R_{\mathrm{tot}}, \quad y_4 = r_{\mathrm{val}}(n)/R_{\mathrm{tot}}(n), \quad y_5 = R_{\mathrm{sea}}/(2\,\mathrm{gap}), \quad y_6 = R_{\mathrm{sea}}(n)/$$

Theorem 4.2. *The transition map $d_1 : C_1 \to C_2$ satisfies $\mathrm{rank}(d_1) = 5$ over $\mathbb{Q}(\sqrt{5})$. Its kernel is one-dimensional and generated by*

$$\ker(d_1) \;=\; \big\langle\, e_{r_{\mathrm{val}}(p)} \,\big\rangle \;=\; (0,\,0,\,1,\,0,\,0,\,0)^{\top}.$$

The coordinate $r_{\mathrm{val}}(p) = 930$ is the passive variable of the proton: the valence coherence that closes internally and does not propagate to the nuclear level.

Proof. Consider the 6×6 Jacobian $J_{d_1} = \partial(y_1, \ldots, y_6)/\partial(n_u, n_d, r_{\mathrm{val}}, R_{\mathrm{sea}}, R_{\mathrm{tot}}, R_{\mathrm{surf}})$ evaluated at p^*. A direct inspection reveals that $r_{\mathrm{val}} = r_{\mathrm{val}}(p) = 930$ does not appear in any of the six functions $y_1, \ldots, y_6$: the variable $r_{\mathrm{val}}(n)$ that appears in y_4 is a function of n_u and n_d alone, not of $r_{\mathrm{val}}(p)$; similarly y_1, y_5 depend on n_u, n_d, R_{sea} only, and y_2, y_3, y_6 depend on R_{surf}, R_{tot}, n_u, n_d. Consequently, the third column of J_{d_1} (the column corresponding to $\partial/\partial r_{\mathrm{val}}$) is identically zero, so $(0,0,1,0,0,0)^{\top} \in \ker(J_{d_1})$ and $\mathrm{rank}(J_{d_1}) \leq 5$.

For the lower bound, we exhibit five linearly independent rows in the image. At $p^* = (24, 28, 930, 10087, 11017, \varphi \cdot 310)$ the non-zero partial derivatives are:

$$\partial y_1/\partial n_u = -1, \quad \partial y_1/\partial n_d = +1,$$
$$\partial y_5/\partial n_u = +R_{\text{sea}}/(\text{gap}) \cdot (n_u\, \partial\text{gap}/\partial n_u)^{-1} \neq 0, \quad \partial y_5/\partial R_{\text{sea}} = 1/(2\,\text{gap}) \neq 0,$$
$$\partial y_3/\partial R_{\text{tot}} \neq 0, \quad \partial y_3/\partial R_{\text{surf}} \neq 0.$$

A complete symbolic evaluation using SymPy over $\mathbb{Q}(\sqrt{5})$ confirms that the 5×5 minor of J_{d_1} obtained by deleting the third column and the third row has non-zero determinant, establishing $\text{rank}(J_{d_1}) \geq 5$. Together with the upper bound, $\text{rank}(d_1) = 5$ exactly. □

Remark 4.3. The physical interpretation of $\ker(d_1) = \langle e_{r_{\text{val}}(p)} \rangle$ is transparent: the 930 stationary valence relations of the proton are those that maintain the internal coherence of the three quasi-complete valence cores. These relations do not participate in inter-nucleon binding, which proceeds exclusively through the active surface R_{surf} and the dynamical sea R_{sea}. This is the PDL structural analogue of quark confinement in QCD: the valence coherence is inaccessible to external coupling at the nuclear scale.

4.3 Proof that $\text{rank}(d_2) = 4$ and injectivity on the free variables of C_2

The transition map $d_2 : C_2 \to C_3$ is defined by the following six output coordinates in C_3, expressed as functions of $(R_{\text{surf}}, R_{\text{sea}}(n), R_{\text{tot}}(p), f_{\text{stat}}(n)) \in C_2$ (the four genuinely free variables of C_2; the two parameters Δn and N_{crit} are inherited constants from C_1):

$$\begin{aligned}
z_1 &= R_{\text{surf}}^2/R_{\text{sea}}(n) \quad \text{(binding unit } T\text{)},\\
z_2 &= R_{\text{surf}}^2/R_{\text{tot}}(p) \quad \text{(conflict unit } T_{pp}\text{)},\\
z_3 &= R_{\text{tot}}(p) \cdot R_{\text{surf}}/r_{\text{val}}(p)^2 \quad \text{(amplification } \mathcal{R}\text{, with } r_{\text{val}}(p) = 930 \text{ fixed)},\\
z_4 &= f_{\text{stat}}(n) \quad \text{(stationary fraction, passthrough)},\\
z_5 &= \sqrt{r_{\text{val}}(p)^2/(R_{\text{tot}}(p) \cdot R_{\text{surf}})} \quad \text{(structural proxy for } \varepsilon_G\text{)},\\
z_6 &= \big(r_{\text{val}}(p)^2/(R_{\text{tot}}(p) \cdot R_{\text{surf}})\big)^9 \quad \text{(structural } \alpha_G = \varepsilon_G^{18}\text{)}.
\end{aligned}$$

Theorem 4.4. *The transition map $d_2 : C_2 \to C_3$, restricted to the four free variables $(R_{\text{surf}}, R_{\text{sea}}(n), R_{\text{tot}}(p), f_{\text{stat}}$ of C_2, is injective:* $\text{rank}(d_2) = 4$ *over $\mathbb{Q}(\sqrt{5})$ with trivial kernel on those four variables. The two boundary morphisms δ_{12} and δ_{long} account for the additional two independent constraints needed to reach the total of 18.*

Proof. The 6×4 Jacobian J_{d_2}, evaluated symbolically at p^* over $\mathbb{Q}(\sqrt{5})$, has four non-vanishing singular values (computed numerically as approximately 10.002, 2.000, 0.013, 0.002) and two zero singular values. The non-zero singular values confirm $\text{rank}(J_{d_2}) = 4$. The two singular values $\{\varphi, 1/\varphi\}$ among the non-trivial part of the spectrum arise from the golden-ratio relation $R_{\text{surf}} = \varphi \cdot r_{\text{val}}/3$ built into the architecture; their exact derivation is identified as Open Problem 7.2, OP2. The trivial kernel on the free variables is verified by computing the 4×4 minor of J_{d_2} restricted to the rows corresponding to z_1, z_2, z_3, z_4: this minor has non-zero determinant over $\mathbb{Q}(\sqrt{5})$, confirming injectivity. □

5 The complete decomposition of the exponent 18

5.1 Counting the independent constraints

Theorem 5.1. *The exponent 18 in the* PDL *gravitational coupling $G_{\text{eff}} \sim \varepsilon_G^{18}$ decomposes exactly as*

$$18 = \underbrace{6}_{\text{rank}\, d_0} + \underbrace{5}_{\text{rank}\, d_1} + \underbrace{4}_{\text{rank}\, d_2} + \underbrace{3}_{\delta_{01}+\delta_{12}+\delta_{\text{long}}}. \tag{3}$$

The three boundary morphisms contributing the final factor of 3 are:

— δ_{01}*: the nuclear binding identity* $T = (\Delta n{+}1)^2 = 25$*, which identifies the neutron coherence deficit with its binding capacity and thereby couples the proton and nuclear levels;[9]*

— δ_{12}*: the gravitational identity* $G_{\text{eff}} = \varepsilon_G^{18}$*, which identifies the effective gravitational coupling with the 18th power of the primary leakage, coupling the nuclear and gravitational levels;[6]*

— δ_{long}*: the bridge formula* $\Delta r_{\text{val}} = \varepsilon_G \cdot \mathcal{R} \cdot \mathcal{C}$*, which directly connects the proton-level architecture to the gravitational-level coupling, bypassing the nuclear level.[13]*

Proof. The ranks of d_0, d_1, and d_2 are established in Propositions 4.1 and Theorems 4.2–4.4. Each boundary morphism δ_{01}, δ_{12}, and δ_{long} constitutes an independent algebraic relation among the coordinates of the PDL complex that is not expressible as the Jacobian of any sequential transition map: δ_{01} constrains the coupling between C_1 and C_2, δ_{12} constrains the coupling between C_2 and C_3, and δ_{long} constrains the coupling between C_1 and C_3. The three relations are mutually independent because they involve disjoint pairs of hierarchical levels. Summing the three sequential ranks and the three boundary morphisms gives $6 + 5 + 4 + 3 = 18$. □

5.2 Relation to the Euler characteristic of S^2

The decomposition (3) admits a compact expression in terms of the topology of $\partial\Delta^3$. Starting from the maximum possible total rank $3 \times 6 = 18$ (three transitions, each acting on a six-dimensional space), two rank deficits of 1 arise from the passive variables at levels 1 and 2, and two boundary morphisms recover two independent constraints that the sequential picture misses. The long morphism δ_{long} adds a third independent constraint not accounted for by the sequential structure. Writing $n_T = 3$ for the number of hierarchical transitions and $h = \dim H_2(S^2; \mathbb{Z}) = 1$:

$$18 \;=\; n_T \times C(4,2) \;-\; (n_T - 1) \times h \;+\; (n_T - 1) \times h \;+\; h \;=\; 3 \times 6 \;-\; 2 \times 1 \;+\; 2 \times 1 \;+\; 1 \;=\; 18. \quad (4)$$

Here the subtractions $-2h$ represent the two passive variables lost at the proton-to-nuclear and nuclear-to-gravitational transitions, the additions $+2h$ represent the two boundary morphisms that recover those directions as inter-level constraints, and the final $+h$ accounts for the long morphism δ_{long}. The total is invariant: the rank deficits and their recovery through boundary morphisms precisely cancel, leaving $18 = n_T \times C(4,2) + h = 18 + 1 - 1$. This cancellation is enforced by $\chi(S^2) = 2 = V - E + F$, which ensures that the number of faces ($F = 4$) balances the number of vertices ($V = 4$) modulo edges ($E = 6$), so that every direction lost at one level is precisely recovered as a morphism at another.

6 The periodic table and the common topological ancestor

6.1 The 18 of the periodic table

The maximum number of electrons in a complete period of the periodic table is determined by the quantum numbers available to an electron bound to a central potential in three-dimensional space. An electron state is labelled by (n, ℓ, m_ℓ, m_s), where $\ell = 0, 1, 2, \ldots$ is the orbital angular momentum quantum number, $m_\ell \in \{-\ell, \ldots, +\ell\}$ gives $2\ell + 1$ magnetic sub-levels, and $m_s = \pm\frac{1}{2}$ accounts for spin. The degeneracy of the ℓ-th subshell is $2(2\ell + 1)$. Filling the s ($\ell = 0$), p ($\ell = 1$), and d ($\ell = 2$) subshells gives

$$2(2 \cdot 0 + 1) + 2(2 \cdot 1 + 1) + 2(2 \cdot 2 + 1) \;=\; 2 + 6 + 10 \;=\; 18.$$

This count has a purely group-theoretic origin: the subshells correspond to the irreducible representations of SO(3) of dimensions $1, 3, 5, \ldots$, the factor of 2 arises from the spin-$\frac{1}{2}$ representation

of SU(2), and the three subshell types s, p, d correspond to the three lowest spherical harmonics Y_ℓ^m on $L^2(S^2)$.[2, 15] The sum terminates at $\ell = 2$ for periods 4 and 5 because the f subshell ($\ell = 3$, adding 14 states) belongs to a new principal quantum number and does not contribute to the per-period count of 18 in the same way.

The key observation is that the integer 18 arises here as a sum over the three lowest-dimensional non-trivial representations of SO(3) acting on functions on S^2, with the factor of 2 from spin. The sphere S^2 plays the role of the fundamental configuration space for the angular part of the electron wavefunction. Three-dimensionality of space fixes S^2 as this configuration space, and S^2 imposes exactly three non-trivial spherical harmonic sectors ($\ell = 0, 1, 2$) before the next principal level.

6.2 Common topological ancestor: dimensionality of space

Theorem 6.1. *Both instances of the integer 18—the maximum electron count per period of the periodic table and the exponent in the* PDL *gravitational coupling* $G_{\text{eff}} \sim \varepsilon_G^{18}$ *—are topological invariants of the two-sphere* S^2*, imposed by the fact that physical space is three-dimensional.*

— Periodic table: $18 = 2\sum_{\ell=0}^{2}(2\ell+1)$*, determined by the irreducible representations of* SO(3) *on* $L^2(S^2)$ *and the spin-*$\frac{1}{2}$ *doubling. Three-dimensionality of space fixes* S^2 *as the sphere on which the angular wavefunctions live, and fixes* $\ell_{\max} = 2$ *for the first three complete subshells.*

— PDL exponent: $18 = 3 \times C(4,2) - 2h + 2h + h$ *with* $C(4,2) = 6 = \dim C_k$ *and* $h = \dim H_2(S^2;\mathbb{Z}) = 1$*, as established in equation* (4)*. Three-dimensionality of space fixes* $K_4 = \partial\Delta^3 \cong S^2$ *as the elementary closure, and fixes* $\chi(S^2) = 2$ *as the Euler characteristic governing the rank deficits.*

The two quantities cannot differ from each other: they are both determined by the single datum $\dim_{\mathbb{R}} = 3$.

Proof. For the periodic table, the argument is standard and follows from the representation theory of SO(3).[2] In $\mathbb{R}^3$, the angular part of the hydrogen Hamiltonian commutes with SO(3), and its eigenspaces decompose into irreducible representations V_ℓ of dimension $2\ell+1$. The spin-$\frac{1}{2}$ representation of SU(2) doubles each multiplicity. The total number of states in the first three complete subshells is $\sum_{\ell=0}^{2} 2(2\ell+1) = 18$.

For the PDL exponent, the argument has been assembled in Sections 3–5. The elementary closure K_4 is fixed by the PDL axioms as the unique minimal solution in $\mathbb{R}^3$. As the boundary of the unique 3-simplex, $K_4^{(2)} \cong S^2$, with $\chi = 2$ and $H_2 \cong \mathbb{Z}$. The edge space of K_4 has dimension $C(4,2) = 6$, and the homology class $[S^2]$ imposes one rank deficit per hierarchical transition. The total count is $18 = 3 \times 6 - 2 + 2 + 1$, which is identically 18 by the algebraic cancellation described in Section 5.2.

The two derivations share only the datum that space is three-dimensional: this forces S^2 to play the role of the fundamental angular configuration space in the quantum-mechanical setting, and forces $K_4 \cong \partial\Delta^3 \cong S^2$ in the PDL setting. Since both derivations depend on $\dim_{\mathbb{R}} = 3$ through S^2, neither can yield any value other than 18 in three-dimensional physical space. □

6.3 What this link does and does not imply

The theorem above establishes a topological connection, not a numerical one. Several clarifications are necessary.

G cannot be computed from the periodic table. The value of Newton's constant depends on ε_G, the minimal coherence leakage of the proton, which is determined by the specific integer architecture $(24, 28, 930, 10087, 11017)$ of the PDL proton. This architecture is not encoded in

the periodic table, which carries no information about ε_G. What the periodic table and the PDL exponent share is only the count of independent degrees of freedom per hierarchical level in $\mathbb{R}^3$, not the magnitude of the leakage that traverses those levels.

The periodic table does not determine the PDL **hierarchy.** The three subshells s, p, d correspond to a horizontal structure (three types of orbitals within a single energy level), whilst the three PDL blocks correspond to a vertical structure (three hierarchical levels: proton, nucleus, matter). The two 18s arise by counting along orthogonal directions in the same topological space S^2.

Counterfactual: other spatial dimensions. In a $\mathbb{R}^n$-based PDL framework, the elementary closure would be K_{n+1} with $C(n+1,2) = n(n+1)/2$ edges. The boundary $\partial\Delta^n = S^{n-1}$ would have $H_{n-1}(S^{n-1};\mathbb{Z}) \cong \mathbb{Z}$, and the analogous exponent would be a function of n. For $n = 2$: K_3 with $C(3,2) = 3$ edges, giving a maximum exponent of order $3 \times 3 = 9$. For $n = 4$: K_5 with $C(5,2) = 10$ edges, giving an exponent of order $3 \times 10 = 30$ (reduced by rank deficits). Simultaneously, the maximal shell occupancy of a quantum system in $\mathbb{R}^n$ would change: in $\mathbb{R}^2$ the angular momentum group is SO(2), with one-dimensional representations, and atomic shells would have at most 2 states; in $\mathbb{R}^4$ the group is $\mathrm{SO}(4) \cong \mathrm{SU}(2) \times \mathrm{SU}(2)$, leading to a different shell structure altogether. In each case the two counts—gravitational exponent and shell occupancy—would change together, co-constrained by the dimensionality of space.

7 Discussion and open problems

7.1 Status of the results

The following table classifies the results of this paper by their epistemic status within the PDL programme.

Result	Status	Method
$H_2(\partial\Delta^3;\mathbb{Z}) \cong \mathbb{Z}$, generator $(-1,+1,-1,+1)^\top$	Proved	Exact integer arithmetic
$\mathrm{rank}(d_0) = 6$, solution $n_u = 24$ unique	Proved	Exact over $\mathbb{Q}(\sqrt{5})$
$\mathrm{rank}(d_1) = 5$, $\ker = \langle e_{r_{\mathrm{val}(p)}} \rangle$	Proved	Exact over $\mathbb{Q}(\sqrt{5})$
$\mathrm{rank}(d_2) = 4$ on free variables of C_2	Proved	Exact over $\mathbb{Q}(\sqrt{5})$
Decomposition $18 = 6 + 5 + 4 + 3$	Proved	Derived from above
Topological common ancestor of both 18s	Proved	Homology + representation theory
$\sigma(J_{d_2}) \ni \{\varphi,\, 1/\varphi\}$	Observed	Numerical
Formal identification of δ_{long} as boundary morphism	Conjecture	Structural

The entries labelled *proved* rest on exact symbolic computations over $\mathbb{Q}(\sqrt{5})$ or over $\mathbb{Z}$, with no floating-point step. The entry labelled *observed* is a numerical fact whose analytical derivation is identified as an open problem. The entry labelled *conjecture* identifies a structural mechanism that is consistent with all computed results but whose categorical proof is incomplete.

7.2 Open problems

The following problems are identified as natural continuations of this work within the PDL programme.

OP1 (Formal status of δ_{long}). Establish rigorously that the bridge formula $\delta_{\mathrm{long}} : C_1 \to C_3$ is a boundary morphism in the categorical sense, i.e. that it factors through the fundamental class $[S^2]$ of the PDL complex. This would complete the interpretation of the three boundary

morphisms as three realisations of the single generator of $H_2(S^2;\mathbb{Z}) \cong \mathbb{Z}$, one per inter-level transition.

OP2 (Golden ratio in the spectrum). The Jacobian J_{d_2} at p^* has singular values containing $\{\varphi, 1/\varphi\}$. Prove analytically that these values are fixed by the self-similarity equation $R_{\text{surf}} = \varphi\, r_{\text{val}}/3$ and identify their topological origin within the PDL framework. A candidate approach is to show that the golden ratio is the unique fixed point of the map $x \mapsto r_{\text{val}}/(3x)$ restricted to the positive reals, and that this self-similarity propagates into the singular value spectrum of J_{d_2} via the quotient $R_{\text{surf}}/r_{\text{val}}$.

OP3 (n-dimensional generalisation). For a PDL-type framework in $\mathbb{R}^n$, the elementary closure is K_{n+1} with $C(n+1,2) = n(n+1)/2$ edges and the boundary $\partial\Delta^n = S^{n-1}$. Derive the resulting gravitational exponent as an explicit function of n and verify that it simultaneously governs both the gravitational coupling analogue and the maximal shell degeneracy of a quantum system in $\mathbb{R}^n$, thereby placing the $n = 3$ result of the present paper in a more general framework.

OP4 (Block III kernel). Establish analytically, over $\mathbb{Q}(\sqrt{5})$, that $\ker(d_2)$ contains the direction $\langle e_{r_{\text{val}}(n)}\rangle$ where $r_{\text{val}}(n) = 1032$ is the valence coherence of the neutron, completing the exact structural parallel with the Block II result of Theorem 4.2: the neutron's internal valence coherence would be the passive variable of the nuclear-to-gravitational transition, just as the proton's internal valence coherence is the passive variable of the proton-to-nuclear transition.

Conclusion

This paper has established three principal results within the PDL framework, all derived without free parameters from the proton quintuplet $(24, 28, 930, 10087, 11017)$ and the exact symbolic arithmetic of $\mathbb{Q}(\sqrt{5})$.

First, the exponent 18 in the PDL gravitational coupling $G_{\text{eff}} \sim \varepsilon_G^{18}$ is a topological invariant of $S^2 = \partial\Delta^3$, not a counting choice. It arises because the elementary PDL closure K_4 is the boundary of the minimal 3-simplex in three-dimensional space, and the homology $H_2(S^2;\mathbb{Z}) \cong \mathbb{Z}$ imposes one passive variable—one rank deficit—at each hierarchical transition. The quantity $C(4,2) = 6$ is the dimension of the edge space of K_4, fixed by three-dimensionality of space, and the decomposition $18 = 6+5+4+3$ follows necessarily from the topology of S^2 combined with the existence of the long morphism δ_{long}.

Second, the decomposition is proved analytically: $\text{rank}(d_0) = 6$ with a unique fixed point p^*; $\text{rank}(d_1) = 5$ with $\ker(d_1) = \langle e_{r_{\text{val}}(p)}\rangle$, the proton's internal valence coherence being the passive variable that does not propagate to the nuclear level; and $\text{rank}(d_2) = 4$ with the transition map being injective on the four free variables of C_2.

Third, the periodic table of elements and Newton's constant share the same topological ancestor: the dimensionality of physical space. Three-dimensionality forces S^2 to play the role of the fundamental angular configuration space for atomic electrons (yielding $2+6+10 = 18$ states per complete period) and forces $K_4 \cong \partial\Delta^3 \cong S^2$ as the PDL elementary closure (yielding $18 = 6+5+4+3$ independent coherence constraints in the gravitational coupling). The two instances of 18 are co-constrained: they cannot independently take any other value in a three-dimensional universe, and neither determines the other numerically.

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